

Aditi Amarnath

(630) 245-3283 | aditia.amarnath@gmail.com | [linkedin.com/in/aditi-amarnath](https://www.linkedin.com/in/aditi-amarnath) | github.com/aditiaamarnath-rgb

EDUCATION

University of Wisconsin–Madison

BS in Computer Science and Statistics

Aug. 2025 – Expected Dec. 2028

Madison, WI

- **GPA:** 3.85
- **Awards:** Dean's List, Margaret E. and Allard Smith Undergraduate Scholarship
- **Relevant Coursework:** Calculus III, Linear Algebra, Data Structures & Object-Oriented Programming (CS300), Algorithms & Software Development (CS400), Computer Engineering, Genetics, Statistics, Statistical Methods in R

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Prof. Andi Wang, Dept. of Industrial Engineering

Sept. 2026 – Present

Madison, WI

- Studying Gaussian processes and optimization and probabilistic network models with application to various engineering problems; reading and summarizing research papers
- Implementing methods in Python and R to run experiments, present progress and results weekly to faculty mentor
- Part of the Undergraduate Research Scholars (URS) program: year-long mentored research with weekly seminars

WORK EXPERIENCE

Cloud Analytics Intern

Hinduja Group Services

June 2026 – Aug. 2026

Warrenville, IL

- Co-authored a technical specification for Databridge, a Microsoft Fabric-native financial data platform, defining the OneLake medallion architecture, Direct Lake semantic layer, capacity tiering, and security across a 17-item decision register
- Built a Power BI semantic model on Microsoft Fabric by flattening a parent-child employee hierarchy into a dimension table and implementing a 7-relationship star schema, then layered an EDA agent on top as an AI proof-of-concept

Private Piano Instructor

The Music and Dance Suite

July 2024 – Aug. 2026

Naperville, IL

- Developed customized lesson plans for 20+ students, using adaptive techniques for neurodiverse and nonverbal learners
- Organized and led performance recitals, improving student involvement and retention

PROJECTS

BluPrint: Personalized Interior Decorator | *React, Three.js, MongoDB, GPT-4o Vision*

May 2026 – Present

- Built a full-stack pipeline where uploaded room images (stored in Cloudinary, linked via MongoDB) are analyzed by GPT-4o Vision to extract style, color palette, and room features, then matched to real IKEA/Wayfair products via Serper.dev/Google Shopping with budget-optimized pricing and a curated catalog for accurate dimensions
- Developed a custom Three.js engine that procedurally renders the room and matched furniture at true scale, with Firebase auth and a React/Vite frontend

Wavelength: Symptom Tracker | *Vite, Tailwind, Firebase, Supabase, GroqCloud, jsPDF*

Sept. 2026 – Present

- Built a full-stack app where users log symptoms by text or voice; recurring symptoms are tagged and graphed over time (Victory), with a calendar view, GroqCloud-generated summaries, and a jsPDF visit brief for doctors
- Added a pre-AI safety check that scans entries for emergency keywords (e.g., chest pain, self-harm) and shows a crisis message instead of calling the model
- Secured data with Firebase auth and Supabase row-level security; supported 20+ languages with right-to-left layouts

Databridge: Financial Data Platform | *Microsoft Fabric, Power BI, OneLake*

June 2026 – Aug. 2026

- Designed 16-feature product roadmap spanning adaptation engine, finance ontology, semantic layer, and intelligence layers, then structured pitch deck with competitive positioning and pricing for enterprise ERP customers

LEADERSHIP AND INVOLVEMENT

Education Chair and Volunteer

Code To Connect

Sept. 2025 – Present

Madison, WI

- Designed Scratch and Java curriculum for meetings and volunteer events, teaching 30+ elementary and middle school students
- Led coding demos and outreach at community fairs to introduce computer science to new audiences

SKILLS

Languages: Java, Python, R, Bash, JavaScript, HTML/CSS, Mathematica, RV32I, C

Tools & Technologies: Git/GitHub, Linux, React, JavaFX, RStudio, Excel, Photoshop, PowerBI, Fabric, Vim

Spoken Languages: English (Fluent), Tamil (Fluent), Spanish (Intermediate)